# Domain Research Note

**Project Title : NER using DEBERTA for recruitment**

**Domain : Natural Language Processing (NLP) → Information Extraction → Recruitment Analytics / HR Technology**

## Introduction

Extracting structured information from unstructured resumes is a challenging task due to the wide variations in resume formats, writing styles, abbreviations, and presentation of candidate information. Traditional rule-based extraction approaches often fail to generalize across diverse resume templates and may produce inaccurate results when handling complex contextual information. As organizations receive a large number of applications for each job opening, manual resume screening becomes time-consuming and inefficient.

Therefore, there is a need for an intelligent automated system capable of extracting relevant candidate information such as skills, education, experience, job roles, and contact details from resume documents and effectively matching them with job requirements. In this work, a transformer-based recruitment framework is proposed that combines spaCy-based Named Entity Recognition, DeBERTa contextual embeddings, semantic similarity techniques, and extractive summarization methods. The extracted candidate information is compared with job descriptions using cosine similarity, Jaccard similarity, and role-matching strategies to rank candidates according to their suitability. This approach aims to improve recruitment efficiency, reduce manual effort, and support accurate candidate selection through AI-driven resume analysis.

## Problem Statement

Extracting structured information from unstructured resumes is challenging due to variations in formatting, abbreviations, and contextual differences across candidate profiles. Traditional rule-based methods are limited in handling these variations effectively. Therefore, there is a

need for an automated NER-based system capable of accurately extracting relevant candidate information and comparing it with job descriptions. In this work, DeBERTa is used for contextual understanding during entity extraction, while cosine similarity and other similaritybased methods are applied to measure the relevance between resumes and job descriptions. BART-based summarization is also used to generate concise resume representations that support efficient candidate evaluation.

## Methodology Overview

The proposed system follows a hybrid NLP pipeline for resume parsing and recruitment automation. Resume documents in PDF format are processed using text extraction techniques, and important entities such as name, education, skills, experience, role, contact number, and email are extracted using spaCy-based Named Entity Recognition supported by DeBERTa contextual embeddings. Extractive summarization is performed using Sentence-BERT embeddings and the TextRank algorithm implemented through NetworkX PageRank. Regular expressions are applied for extracting structured attributes such as contact numbers and email addresses. A pretrained DeBERTa model is utilized to generate contextual embeddings for entity extraction and semantic analysis.

## Resume Ranking Strategy and Significance

After extracting candidate information, resumes are compared with job descriptions using cosine similarity for contextual matching, Jaccard similarity for skill comparison, and sequence matching for role alignment. A weighted composite scoring mechanism combines skill match, experience match, role match, and contextual similarity to rank resumes based on suitability. This approach improves recruitment efficiency by reducing manual screening effort and supports accurate identification of relevant candidates using transformer-based entity extraction and similarity-based ranking techniques.